Ben Altangerel

+44 7984 611311 | bayasal@hotmail.co.uk | linkedin.com/in/bayasalaltangerel | London

Professional Summary

Aerospace Engineering graduate (MEng, 2:1) with experience building machine learning models in Python and SQL, from forecasting customer lifetime value to predicting Premier League results, plus over five years in hospitality.

Technical Skills

Programming: Python (pandas, NumPy, SciPy, scikit-learn, LightGBM, XGBoost, Matplotlib, pytest), SQL (SQLite), C#, MATLAB
Data Science: Data Cleaning, Feature Engineering, Regression and Ensemble models, Neural Networks, Customer Lifetime Value Analysis
Tools: Excel, VS Code, Claude Code, ParaView, ANSYS Fluent

Projects

Premier League Match Outcome Prediction with Machine Learning Aug. 2026 – Oct. 2026
Independent Project *Manchester*

- Built a random forest pipeline that matched bookmakers' 49.5% accuracy on 380 unseen Premier League matches (+6.9 points over baseline), trained on 2,330 matches with a time-based split.
- Improved logistic regression accuracy from 46.1% to 49.5% and log loss by 3% by expanding training data from 1 to 5 seasons; a learning curve showed gains level off after 2–4 seasons.
- Automated weekly forecasts producing next-round predictions and 12 charts in 15 seconds, scoring every prediction against bookmakers' odds and plotting a form guide of all 20 teams' last 5 results and predicted next match.

Inventory & Financial Tracking System Jun. 2026
Independent Project *Manchester*

- Built a Python and SQLite inventory system producing weekly revenue, cost and profit figures that matched the business's in-house records to 91% accuracy, by automating stock deduction against defined portion levels and calculating weighted-average supplier costs from logged deliveries.
- Enabled week-on-week performance comparison and identification of stock discrepancies by exporting financial records to Excel for charting and reconciling system stock levels against physical counts.

Predicting Customer Lifetime Value with Machine Learning Aug. 2025 – Nov. 2025
Independent Project *Sheffield*

- Built a reproducible 31-feature modelling dataset from 9,134 raw insurance records by designing a tested Python pipeline for data cleaning, categorical encoding and feature engineering.
- Took prediction accuracy from R^2 0.16 with a simple linear model to 0.71 with tree-based models, by building and comparing five approaches, from Random Forest and gradient boosting to a stacked ensemble, and testing each one fairly with 5-fold cross-validation.
- Found and fixed a data leak that had inflated model accuracy from R^2 0.71 to 0.999, by tracing how the answer slipped into the model's inputs and adding checks to stop it recurring.

Aerospace Design III – Transitioning Fixed-Wing UAS Sep. 2023 – Jun. 2024
Project Team Lead *Sheffield*

- Delivered a drone that met all of Aeromap Ltd's requirements, finishing at £519.59 against an £800 budget, by leading a five-person team and managing schedules, design reviews and milestones.
- Kept the project on schedule and on budget, with clear visibility of spending and risks, by maintaining the bill of materials, budget tracker, risk assessments and weekly progress reports.
- Improved aerodynamic efficiency by over 16% by training a neural network on 50 simulation results and running a Python optimisation algorithm to find the best-performing wing design.

Education

The University of Sheffield Sheffield
Master of Engineering in Aerospace Engineering *Upper Second-Class Honours (2:1)*

Loreto College Manchester
A-Levels: Law (A), Physics (B), Mathematics (B) *Sep. 2019 – Jun. 2021*

Work Experience

Waiter and Front of House Nov. 2024 – Present
BOX (ARC Inspirations) *Manchester & Sheffield*

- Received the most positive customer reviews on the team during the World Cup (93) by proactively handling queries and prioritising orders at peak service.
- Kept service accurate and on time during busy periods by communicating clearly with managers, kitchen staff and co-workers, relaying changes and flagging issues early.

Supervisor Jul. 2022 – Oct. 2024
Fuku *Manchester*

- Cut the restaurant's supply costs by 12% by approaching potential suppliers, comparing their prices and negotiating better rates.
- Kept the restaurant fully stocked for every service by carrying out regular stock counts and keeping inventory records accurate and up to date.

Bartender and Back of House Jun. 2021 – Jul. 2022
Turtle Bay *Manchester & Sheffield*

- Recommended, prepared and served drinks and cocktails to the guests, as requested, with pace and accuracy.
- Maintained the cleanliness of the bar and equipment to ensure all drinks are served to the highest quality.

Career Break – COVID-19 Restrictions Apr. 2020 – May 2021
Full-time A-Level Student *Manchester*

- Out of work during UK COVID-19 restrictions, from the first national lockdown on 23 March 2020 until indoor hospitality reopened on 17 May 2021, while completing A-levels at Loreto College.

Retail Assistant Apr. 2019 – Mar. 2020
Topshop and Topman *Manchester*

- Operated tills and assisted customers on the shop floor, delivering friendly and attentive customer service.
- Showcased attention to detail and a proactive work ethic by maintaining store cleanliness and organising inventory.

Laboratory Assistant May 2019
University of Manchester Biology Research *Manchester*

- Logged 60 experimental results into the lab's spreadsheet on the day they were produced, checking each against the paper record and correcting 3 transcription errors.
- Counted and updated all 120 items on the lab inventory by midweek and flagged 5 below minimum stock with the quantities needed, so the order was placed before the next experiment.
- Acted as first point of contact for around 10 calls a day and managed 6 lab bookings and 4 staff meetings, resolving 1 double booking before it affected an experiment.

Extra-Curricular Activities

Football

- Organiser and treasurer for a workplace football team, running weekly matches and finances. 2025 – Present
- Won a year-long 5-a-side football knockout competition organised by the University of Sheffield. 2023 – 2024

Volunteer Apr. 2026 – Present
Hackspace Manchester *Manchester*

- Volunteer at a community makerspace, running training sessions and tours across laser cutting, 3D printing, electronics and metalwork workshops.